

Social Computing

Unit 2: Network Measures

April 2023

Q3) a) What is Centrality? Explain Degree Centrality with example.

Ans. In **social computing**, centrality is used to identify how important or influential a node (person, account, or entity) is within a **social network**.

1. Centrality

Centrality is a measure used in **social network analysis (SNA)** to determine the importance or influence of a node within a network.

- In a social network, **nodes** represent users and **edges** represent relationships (friendship, followers, communication, etc.).
- Centrality helps identify **key actors, influencers, or important connectors** in the network.

Example:

On a platform like Twitter or Facebook, users who interact with many others or have many followers usually have **higher centrality**, meaning they have greater influence in spreading information.

Uses of Centrality in Social Computing

- Identifying influencers in social media
- Studying information flow in networks
- Detecting important communities or leaders
- Marketing and recommendation systems

2. Degree Centrality

Degree Centrality measures the importance of a node based on the **number of direct connections it has with other nodes**.

Definition:

Degree centrality is the **count of how many links (edges) a node has with other nodes in the network**.

Formula

$$C_D(v) = \deg(v)$$

Where:

- $C_D(v)$ = Degree centrality of node (v)
- $\deg(v)$ = Number of connections of node (v)

In normalized form:

$$C_D(v) = \frac{\deg(v)}{n - 1}$$

Where **n** is the total number of nodes in the network.

Example

Consider a simple social network:

- A connected to B, C, D
- B connected to A, C
- C connected to A, B
- D connected to A

Node	Connections	Degree Centrality
A	B, C, D	3
B	A, C	2
C	A, B	2
D	A	1

Explanation:

- Node **A** has the highest degree centrality (3 connections).
- This means **A is the most connected and influential user** in this small network.

Types of Degree Centrality

1. **In-Degree Centrality** – Number of incoming connections (e.g., followers on Instagram).
2. **Out-Degree Centrality** – Number of outgoing connections (e.g., accounts a user follows).

✓ Summary:

Centrality measures the importance of nodes in a social network. **Degree centrality** specifically measures influence based on the **number of direct connections a node has**, helping identify the most connected or active users in social computing systems.

Q3) b) Explain network relationship using transitivity and transitivity linking.

Ans. In **social computing and social network analysis**, network relationships explain how individuals (nodes) are connected through relationships (edges). Two important concepts used to understand these relationships are **Transitivity** and **Transitivity Linking**.

1. Transitivity

Transitivity refers to the tendency of people who share a common connection to also become connected with each other.

It follows the principle:

“A friend of my friend is likely to become my friend.”

If:

- Person **A** is connected to **B**
- Person **B** is connected to **C**

Then there is a high probability that:

- **A will also connect with C**

This forms a **triangle relationship** in the network.

Example

On social media platforms like Facebook or Instagram:

- A is friends with B
- B is friends with C
- The platform may suggest that A and C become friends.

This creates a **closed triangle network**, showing **transitivity**.

Importance of Transitivity

- Shows **strong social ties**
- Helps identify **communities or clusters**
- Improves **friend recommendation systems**

2. Transitivity Linking

Transitivity linking refers to the **process of forming a new connection between two nodes because they share a common neighbor**.

It describes how networks **grow and become more connected over time**.

Example

Consider a network:

- $A \rightarrow B$
- $B \rightarrow C$

Through **transitivity linking**, a new link may form:

- $A \rightarrow C$

Now the network becomes:

- $A \leftrightarrow B$
- $B \leftrightarrow C$
- $A \leftrightarrow C$

This creates a **fully connected triangle**.

On platforms like LinkedIn, this appears as **“People You May Know”** recommendations based on mutual connections.

Difference Between Transitivity and Transitivity Linking

Aspect	Transitivity	Transitivity Linking
Meaning	Property of a network where connected friends tend to become connected	Process of creating new connections based on mutual neighbors
Focus	Existing relationship pattern	Formation of new relationships
Result	Closed triangular network	Network expansion

✓ Summary:

- **Transitivity** describes the tendency of networks to form **triangles of relationships**.
- **Transitivity linking** is the **process that creates those new connections** based on shared contacts, strengthening the overall network structure.

Q3) c) Explain Page Ranking algorithm.

Ans. **Page Ranking Algorithm**

The **Page Ranking algorithm (PageRank)** is a method used to measure the **importance of web pages in a network of pages**. It was originally developed by Larry Page and Sergey Brin at Google to rank webpages in search results.

The algorithm works on the idea that **a webpage is important if many other important pages link to it**.

Basic Idea

PageRank treats the **web as a directed graph**:

- **Nodes** → Web pages
- **Edges (links)** → Hyperlinks between pages

If a page receives many links from other pages, it is considered **more important**.

A link from one page to another acts like a **vote of importance**.

PageRank Formula

$$PR(A) = (1 - d) + d \left(\frac{PR(T1)}{C(T1)} + \frac{PR(T2)}{C(T2)} + \dots + \frac{PR(Tn)}{C(Tn)} \right)$$

Where:

- **PR(A)** = PageRank of page A
- **PR(T1, T2 ... Tn)** = PageRank of pages linking to page A
- **C(T)** = Number of outgoing links from page T
- **d** = Damping factor (usually **0.85**)

The **damping factor** represents the probability that a user continues clicking links rather than starting a new search.

Working of the PageRank Algorithm

1. Initialize ranks

Each page is given an initial PageRank value.

2. Distribute rank through links

A page shares its rank with the pages it links to.

3. Iterative calculation

The algorithm repeatedly recalculates PageRank values.

4. Convergence

After several iterations, the values stabilize and represent the final ranking.

Example

Consider three pages: **A, B, and C**

- A links to B
- B links to C
- C links to A and B

In this network:

- Page **B** receives links from **A and C**, so its rank becomes higher.
- Page **C** receives only one link, so it may have a lower rank.

Thus, **pages with more incoming links from important pages get higher PageRank.**

Applications of PageRank

- Ranking search engine results in Google Search
- Measuring influence in **social networks**
- Identifying **important nodes in network analysis**
- Recommendation systems and information retrieval

Summary:

The **PageRank algorithm** evaluates the importance of webpages by analyzing the **number and quality of incoming links**. Pages linked by many high-ranked pages receive a **higher rank**, helping search engines display the most relevant pages first.

Q4) a) Explain Structural Equivalence and Regular Equivalence.

Ans. In **social network analysis (SNA)** and **social computing**, **equivalence** refers to how similar two nodes (people, organizations, etc.) are based on their **relationships with other nodes** in the network. Two important types are **Structural Equivalence** and **Regular Equivalence**.

1. Structural Equivalence

Structural Equivalence occurs when **two nodes have identical relationships with the same set of nodes** in a network.

This means:

- They are connected to **exactly the same people**

- Their **network position is identical**

If node **A** and node **B** both connect to the **same nodes**, they are structurally equivalent.

Example

Consider a network:

- $A \rightarrow C$
- $A \rightarrow D$
- $B \rightarrow C$
- $B \rightarrow D$

Here:

- **A and B connect to the same nodes (C and D)**
- Therefore, **A and B are structurally equivalent**

Interpretation:

A and B play the **same role in the network** because their connections are identical.

Characteristics

- Very **strict definition of similarity**
- Nodes must have **exactly the same neighbors**
- Used to identify **nodes with identical positions**

Example in Social Media

On platforms like Facebook, two users who have **exactly the same friends** can be considered **structurally equivalent**.

2. Regular Equivalence

Regular Equivalence occurs when **two nodes have similar roles in the network, even if they are connected to different nodes**.

In this case:

- Nodes are not required to connect to the **same individuals**
- They just connect to **similar types of nodes**

Example

Consider a network:

- Teacher 1 → Students A, B
- Teacher 2 → Students C, D

Here:

- Teachers connect to students
- Students connect to teachers

Even though the teachers connect to **different students**, **Teacher 1 and Teacher 2 are regularly equivalent** because they perform the **same role**.

Characteristics

- **More flexible than structural equivalence**
- Focuses on **similar roles or functions**
- Widely used in **role analysis**

Example in Social Media

On LinkedIn, different **managers** supervising different teams may be **regularly equivalent** because they play the **same managerial role**, even though their connections differ.

Difference Between Structural and Regular Equivalence

Aspect	Structural Equivalence	Regular Equivalence
Definition	Nodes have identical connections to the same nodes	Nodes have similar roles but not necessarily same connections
Strictness	Very strict	More flexible
Focus	Same neighbors	Same role in the network

Example	Two users with identical friend lists	Two managers managing different teams
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✓ Summary:

- **Structural Equivalence** means two nodes have **exactly the same connections**.
- **Regular Equivalence** means two nodes play **similar roles in the network**, even if they connect to different nodes.

Both concepts help analyze **roles, positions, and patterns of relationships in social networks**.

Q4) b) Explain Balance & Status theory with example.

Ans. In **social computing and social network analysis**, **Balance Theory** and **Status Theory** explain how relationships between individuals behave and evolve in a social network. These theories are mainly used to study **friendships, trust, and conflicts** among people.

1. Balance Theory

Balance Theory was proposed by Fritz Heider. It explains how **relationships among three individuals tend to move toward a balanced or stable state**.

It is based on the idea that people prefer **consistent and harmonious relationships**.

Basic Concept

Relationships are represented using:

- **Positive (+)** → Friendship / liking
- **Negative (-)** → Dislike / hostility

Balance theory usually studies **triads (three people)**.

Balanced and Unbalanced Triads

Relationship Pattern	Balanced / Unbalanced
+ + +	Balanced
+ - -	Balanced
+ + -	Unbalanced
- - -	Unbalanced

Example

Suppose three people: **A, B, and C**

Case 1 (Balanced):

- A likes B (+)
- B likes C (+)
- A likes C (+)

This is balanced because **everyone likes each other**.

Case 2 (Balanced):

- A likes B (+)
- B dislikes C (-)
- A dislikes C (-)

This is balanced because **“the enemy of my friend is my enemy.”**

Case 3 (Unbalanced):

- A likes B (+)
- B likes C (+)
- A dislikes C (-)

This creates **psychological tension**, and the network tends to change to become balanced.

Applications

- Studying **friend groups and conflicts**
- Understanding **community formation**

- Predicting **changes in social relationships**

2. Status Theory

Status Theory explains relationships in **directed networks** where links represent **status or hierarchy**.

It suggests that:

- Individuals tend to form links toward **higher-status people**.
- The direction of relationships indicates **relative status**.

Basic Idea

If:

- **A → B** (A follows or supports B)
- It means **B has higher status than A**.

Example

Consider three users in a social network:

- A follows B
- B follows C

Here:

- C has the **highest status**
- B has **medium status**
- A has **lower status**

In platforms like Twitter or Instagram, many users follow **celebrities or influencers**, showing **higher status**.

Characteristics

- Works with **directed networks**

- Focuses on **hierarchy and influence**
- Helps identify **leaders or high-status nodes**

Difference Between Balance Theory and Status Theory

Aspect	Balance Theory	Status Theory
Focus	Harmony in relationships	Hierarchy in relationships
Network Type	Undirected relationships	Directed relationships
Structure	Triads (groups of three)	Status ranking among nodes
Example	Friend/enemy relations	Followers and influence

✓ Summary:

- **Balance Theory** explains how relationships tend to become **stable and balanced** in small groups.
- **Status Theory** explains how **hierarchy and influence** form in directed social networks.

Both theories help understand **social behavior and relationship patterns in social computing networks**.

Q4) c) Explain Betweenness Centrality and Closeness Centrality.

Ans. In **social network analysis (SNA)** and **social computing**, **centrality measures** are used to identify the importance or influence of nodes (users) in a network. Two important measures are **Betweenness Centrality** and **Closeness Centrality**.

1. Betweenness Centrality

Betweenness Centrality measures how often a node lies on the **shortest path between other nodes** in a network.

It indicates how much a node acts as a **bridge or intermediary** between other nodes.

A node with high betweenness centrality can **control or influence the flow of information** in the network.

Formula

$$C_B(v) = \sum \frac{\sigma_{st}(v)}{\sigma_{st}}$$

Where:

- $(C_B(v))$ = Betweenness centrality of node (v)
- (σ_{st}) = Total number of shortest paths between node (s) and (t)
- $(\sigma_{st}(v))$ = Number of those paths that pass through node (v)

Example

Consider a network:

A — B — C — D

Here:

- Node **B** and **C** lie between other nodes.
- Many shortest paths pass through **B** and **C**.

So **B and C have higher betweenness centrality** because they connect different parts of the network.

Importance

- Identifies **gatekeepers or brokers**
- Helps analyze **information flow**
- Useful in **communication and social media networks**

Example: On Twitter, a user connecting two different communities may have **high betweenness centrality**.

2. Closeness Centrality

Closeness Centrality measures how **close a node is to all other nodes in the network**.

It indicates how quickly a node can **reach other nodes**.

A node with high closeness centrality can **spread information faster**.

Formula

$$\left[C_c(v) = \frac{1}{\sum d(v,u)} \right]$$

Where:

- $(C_c(v))$ = Closeness centrality of node (v)
- $(d(v,u))$ = Shortest distance between node (v) and node (u)

Example

Consider this network:

A — B — C — D — E

- Node **C** is located in the **middle of the network**.
- It has the **shortest total distance to all other nodes**.

Therefore, **C has the highest closeness centrality**.

Importance

- Identifies nodes that can **spread information quickly**
- Useful for **communication efficiency**
- Helps find **central users in social networks**

Example: A highly connected user on Facebook who can quickly reach many friends has **high closeness centrality**.

Difference Between Betweenness and Closeness Centrality

Aspect	Betweenness Centrality	Closeness Centrality
Meaning	Measures how often a node lies between other nodes	Measures how close a node is to all other nodes
Role	Acts as a bridge or mediator	Acts as a fast communicator
Focus	Control of information flow	Speed of reaching others
Example	Connector between communities	Node at the center of network

✓ Summary:

- **Betweenness Centrality** identifies nodes that act as **bridges connecting different parts of a network**.
- **Closeness Centrality** identifies nodes that can **reach all other nodes quickly** in a network.

Both measures help analyze **influence and communication efficiency in social computing networks**.

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Q3) b) Explain transitivity and reciprocity with equations.

Ans. In **social network analysis (SNA)** and **social computing**, **transitivity** and **reciprocity** are measures used to understand **relationship patterns among nodes (users)** in a network.

1. Transitivity

Transitivity refers to the tendency that **if node A is connected to B, and B is connected to C, then A is likely to be connected to C**.

It follows the concept:

“Friend of my friend is also my friend.”

This usually forms a **triangle relationship** among three nodes.

Transitivity Equation

$$T = \frac{\text{Number of Closed Triplets}}{\text{Total Number of Triplets}}$$

Where:

- **Closed triplet** → Three nodes where all are connected (forming a triangle)
- **Triplet** → Three nodes connected by two or three links

Example

Suppose in a network:

- $A \rightarrow B$
- $B \rightarrow C$
- $A \rightarrow C$

Here all three nodes are connected, forming a **triangle**, so the network shows **high transitivity**.

If there were only:

- $A \rightarrow B$
- $B \rightarrow C$

Then it would be an **open triplet** (no transitivity).

Importance

- Detects **community structures**
- Shows **clustering in social networks**
- Used in **friend recommendation systems**

2. Reciprocity

Reciprocity measures the extent to which **relationships in a directed network are mutual**.

It shows whether two nodes **have bidirectional connections**.

For example:

- If **A follows B** and **B also follows A**, the relationship is **reciprocal**.

Reciprocity Equation

$$R = \frac{\text{Number of Mutual Links}}{\text{Total Number of Links}}$$

Where:

- **Mutual links** → Two nodes connected in both directions
- **Total links** → All directed edges in the network

Example

Consider these connections:

- $A \rightarrow B$
- $B \rightarrow A$
- $C \rightarrow A$

Here:

- A and B have a **mutual connection** (reciprocal).
- $C \rightarrow A$ is **not reciprocal**.

So reciprocity measures the **proportion of mutual relationships**.

Importance

- Indicates **strength of relationships**
- Measures **mutual interaction**

- Useful in analyzing **communication networks and social media**

Difference Between Transitivity and Reciprocity

Aspect	Transitivity	Reciprocity
Meaning	Measures triangular relationships among three nodes	Measures mutual relationships between two nodes
Network Pattern	A-B-C triangle	$A \leftrightarrow B$ mutual link
Focus	Group relationship structure	Pairwise relationship
Use	Community detection	Interaction strength

✅ Summary:

- **Transitivity** measures the tendency of networks to form **triangular connections among three nodes**.
- **Reciprocity** measures **mutual relationships between two nodes** in directed networks.

Both help analyze **relationship patterns and interaction behavior in social networks**.

Q4) a) What is Centrality? Explain types of centrality.

Ans. In **social computing and social network analysis (SNA)**, **centrality** measures the **importance, influence, or position of a node (user) in a network**. Different types of centrality help identify influential users, connectors, and information spreaders in social networks.

Types of Centrality

1. Degree Centrality

Degree centrality measures the **number of direct connections a node has** with other nodes in the network.

Equation

$$C_D(v) = \deg(v)$$

or normalized form

$$C_D(v) = \frac{\deg(v)}{n - 1}$$

Where

- $\deg(v)$ = number of links connected to node (v)
- n = total number of nodes

Example:

If a user has many friends on Facebook, that user has **high degree centrality**.

Types

- **In-degree** – number of incoming links
- **Out-degree** – number of outgoing links

2. Betweenness Centrality

Betweenness centrality measures how often a node lies on the **shortest path between other nodes**. It shows how much a node acts as a **bridge or mediator**.

Equation

$$C_B(v) = \sum \frac{\sigma_{st}(v)}{\sigma_{st}}$$

Where

- (σ_{st}) = total shortest paths between nodes (s) and (t)
- $\sigma_{st}(v)$ = number of those paths passing through node (v)

Example:

A user connecting two different groups on Twitter may have **high betweenness centrality**.

3. Closeness Centrality

Closeness centrality measures how **close a node is to all other nodes in the network**.

A node with high closeness can **reach other nodes quickly**.

Equation

$$C_c(v) = \frac{1}{\sum d(v, u)}$$

Where

- $d(v, u)$ = shortest distance between node (v) and node (u)

Example:

A person located in the **center of a network** who can quickly communicate with everyone has **high closeness centrality**.

4. Eigenvector Centrality

Eigenvector centrality measures the importance of a node based on **connections with other important nodes**.

A node connected to highly influential nodes will also have **high eigenvector centrality**.

Equation

$$C_E(v) = \frac{1}{\lambda} \sum A_{vu} C_E(u)$$

Where

- A_{vu} = adjacency matrix value
- $C_E(u)$ = centrality of neighboring nodes
- λ = constant

Example:

A user followed by many influential people on Instagram will have **high eigenvector centrality**.

Summary Table

Type of Centrality	Meaning	Key Idea
Degree Centrality	Number of direct connections	Most connected node
Betweenness Centrality	Node acting as a bridge	Controls information flow
Closeness Centrality	Distance from other nodes	Fast communication
Eigenvector Centrality	Influence based on influential neighbors	Importance of connections

✓ Conclusion:

Centrality measures help identify **important nodes in a network**. Degree centrality shows **connections**, betweenness shows **bridging nodes**, closeness shows **communication efficiency**, and eigenvector centrality measures **influence through important connections**.

Q4) b) What is information cascade? Explain diffusion in cascades.

Ans. **Information Cascade**

An **information cascade** occurs when individuals **make decisions based on the actions or decisions of others rather than their own information**. In social networks, once a few people adopt an idea or behavior, others follow them, creating a **chain reaction of adoption**.

In simple terms, people **ignore their own knowledge and imitate earlier decisions**, causing information or behavior to spread rapidly through a network.

Example

Suppose a new mobile app becomes popular:

1. Person **A** downloads the app.
2. Person **B** sees A using it and also downloads it.
3. Person **C** sees both A and B using it and follows them.

Now many others begin downloading it just because **others are doing it**, creating an **information cascade**.

This often happens on platforms like Instagram or Twitter when posts, trends, or hashtags go viral.

Characteristics of Information Cascade

- Decisions depend on **observing others' actions**
- Can cause **rapid spread of trends or misinformation**
- Often leads to **viral content**

Diffusion in Cascades

Diffusion in cascades refers to the **process by which information, behaviors, or innovations spread from one node to another in a network**, creating a cascade effect.

It explains **how information propagates step by step through connected individuals**.

Diffusion Process

1. Initial adopter (seed node)

The first person adopts or shares the information.

2. Early adopters

People directly connected to the first adopter receive and adopt the information.

3. Propagation

These adopters influence their neighbors.

4. Cascade growth

The process continues until many nodes adopt the information.

Example of Cascade Diffusion

Consider a social network:

$A \rightarrow B \rightarrow C \rightarrow D \rightarrow E$

- **A shares a post**
- **B sees and shares it**
- **C sees B's post and shares it**
- The information spreads through the network like a **cascade**

This is how **viral videos or memes spread online**.

Types of Cascades (Common Patterns)

1. Broadcast Cascade

One user shares information with many followers.

2. Chain Cascade

Information spreads sequentially from one user to another.

3. Tree Cascade

One user influences multiple users, who then influence others, forming a tree-like structure.

Importance in Social Computing

Information cascades and diffusion help in:

- Understanding **viral marketing**
- Studying **spread of rumors or fake news**
- Predicting **trend formation in social media**
- Designing **recommendation systems**

✓ Summary:

An **information cascade** occurs when people adopt decisions based on **others' actions**, creating a chain reaction. **Diffusion in cascades** describes the **process of spreading information through a network step by step**, leading to viral spread in social media and online communities.

Pre Insem

Q3) a) What is Individual Behavior Modeling? Explain Social Community Structure.

Ans. Individual Behavior Modeling

Individual Behavior Modeling (IBM) is the process of **analyzing and predicting the behavior of individual users in a social network** based on their actions, preferences, and interactions.

It studies how a person **acts, reacts, and makes decisions** within a social system.

Key Factors in Individual Behavior Modeling

1. **User preferences** – interests, likes, and dislikes.
2. **Past behavior** – previous posts, comments, and activities.
3. **Social influence** – effect of friends or followers.
4. **Interaction patterns** – frequency of communication with others.

Example

On platforms like Instagram or Facebook:

- If a user frequently likes **travel posts**, the system predicts that the user may be interested in **tourism-related content**.
- This behavior modeling helps recommend **similar posts, ads, or pages**.

Applications

- Personalized **recommendation systems**
- **Targeted advertising**
- Predicting **user engagement**
- Studying **information diffusion**

Social Community Structure

Social Community Structure refers to the **organization of a network into groups of nodes (communities) that are more densely connected to each other than to the rest of the network**.

A **community** is a group of individuals who interact **more frequently with each other than with outsiders**.

Characteristics of Community Structure

1. **Dense internal connections** within the community
2. **Sparse connections** between different communities
3. Members share **common interests, activities, or relationships**

Example

In a social network:

- Group 1: College friends
- Group 2: Family members
- Group 3: Work colleagues

Each group forms a **community** because members interact more within the group.

Platforms like LinkedIn often show communities such as **professional groups or industry networks**.

Types of Communities

1. **Interest-based communities** – people with similar hobbies or interests
2. **Location-based communities** – people from the same place
3. **Professional communities** – people from the same profession
4. **Online communities** – groups formed in social media platforms

Importance of Community Structure

- Helps understand **social interactions**
- Useful for **marketing and recommendation systems**
- Helps detect **influential groups**
- Assists in **information spreading analysis**

Summary:

- **Individual Behavior Modeling** studies and predicts how individuals behave in social networks based on their actions and interactions.
- **Social Community Structure** describes how networks form **groups or clusters of closely connected individuals** sharing common interests or relationships.

Q3) c) Write major components of Behavior Analysis Methodology.

Ans. In **social computing**, **Behavior Analysis Methodology** refers to the systematic process used to **study, analyze, and understand the behavior of individuals or groups in social networks**. It helps researchers identify **patterns of interaction, influence, and decision-making**.

The **major components of Behavior Analysis Methodology** are:

1. Data Collection

This is the **first step**, where behavioral data of users is gathered from different sources.

Sources include:

- Social media activities
- User profiles
- Posts, likes, comments, and shares

Platforms like Facebook or Twitter provide large amounts of data for analysis.

Purpose:

To collect relevant information about **user actions and interactions**.

2. Data Preprocessing

After data collection, the data is **cleaned and organized** to make it suitable for analysis.

Activities include:

- Removing duplicate or irrelevant data
- Handling missing values
- Formatting and structuring the data

Purpose:

To ensure **accuracy and quality of data** before analysis.

3. Feature Extraction

In this stage, important **behavioral features** are identified from the data.

Examples of features:

- Number of posts
- Frequency of interactions
- Friend or follower connections
- User engagement levels

Purpose:

To identify meaningful variables that represent **user behavior patterns**.

4. Behavior Modeling

Behavior modeling involves **building models that represent how users behave** in the network.

These models may include:

- Statistical models
- Machine learning models
- Network analysis models

Purpose:

To **understand and predict user actions**.

5. Behavior Analysis and Interpretation

In this step, the results of the model are analyzed to **discover patterns and insights**.

Examples:

- Identifying influential users
- Understanding community behavior
- Detecting trends or anomalies

Purpose:

To interpret behavioral patterns and understand **social dynamics**.

6. Evaluation and Validation

The final step is to **evaluate the accuracy and effectiveness of the behavior model**.

Methods include:

- Performance testing
- Comparing predictions with real data
- Model validation

Purpose:

To ensure the model **correctly represents user behavior**.

 Summary:

The major components of Behavior Analysis Methodology include:

1. Data Collection
2. Data Preprocessing
3. Feature Extraction
4. Behavior Modeling
5. Behavior Analysis and Interpretation
6. Evaluation and Validation

These steps help researchers **analyze and predict user behavior in social networks**.

Q4) a) What is Centrality? Explain Degree Centrality and Katz Centrality with example.

Ans. **Katz Centrality**

Katz Centrality is a measure used in **social network analysis (SNA)** to determine the **influence of a node in a network** by considering both **direct and indirect connections**.

Unlike degree centrality (which counts only direct links), Katz centrality also considers **connections through other nodes**, but gives **less weight to distant nodes**.

Basic Idea

A node is considered **important if it is connected to many nodes**, and also if it is connected to **other important nodes**.

So Katz centrality counts:

- **Direct connections**
- **Indirect connections (paths through other nodes)**

But longer paths contribute **less influence**.

Katz Centrality Equation

$$C_K(v) = \alpha \sum A_{vu} C_K(u) + \beta$$

Where:

- $C_K(v)$ = Katz centrality of node (v)
- A_{vu} = adjacency matrix (connection between nodes)
- α = attenuation factor (controls influence of neighbors)
- β = constant initial centrality value

Another representation:

$$C_K = (I - \alpha A)^{-1} \beta$$

Where:

- A = adjacency matrix
- I = identity matrix

Example

Consider a network:

- A connected to B
- B connected to C and D
- C connected to D

Here:

- **B** receives connections from multiple nodes.
- **C and D** also connect through B.

Even if node **A** has only one direct connection, it may still gain influence because it connects to **B**, which is an important node.

Thus:

- **B gets highest Katz centrality**
- Nodes connected to influential nodes also gain **higher centrality values**.

Key Characteristics

- Considers **both direct and indirect influence**
- Uses a **decay factor** so distant nodes contribute less
- Works well in **large networks**

Applications

- Identifying influential users in social networks
- Studying information spread in networks
- Ranking nodes in communication systems
- Used in recommendation systems

Example: On Twitter, a user followed by influential users may have **high Katz centrality** even if their direct followers are few.

✓ Summary:

Katz centrality measures node importance by considering **both direct and indirect connections**, giving smaller weights to distant connections. It helps identify **influential nodes in complex social networks**.

Group Centrality

Ans. **Group Centrality**

Group Centrality is a concept in **social network analysis (SNA)** that measures the **importance or influence of a group of nodes rather than a single node** in a network.

Instead of evaluating one individual, group centrality evaluates **how influential a set of nodes (a group or community) is within the network**.

A group with high centrality can **control communication, influence other nodes, or connect different parts of the network**.

Definition

Group Centrality is the measure of how well a **group of nodes is positioned to influence, control, or access other nodes in the network**.

It is calculated using extensions of different centrality measures such as:

- Degree centrality
- Betweenness centrality
- Closeness centrality

Types of Group Centrality

1. Group Degree Centrality

It measures **how many nodes outside the group are connected to the group members**.

Equation

$$C_D(G) = |\{v \notin G : \exists u \in G \text{ such that } (u, v) \in E\}|$$

Where:

- G = group of nodes
- E = edges in the network

Example:

If a team of users in a network connects to many other users, that team has **high group degree centrality**.

2. Group Betweenness Centrality

It measures how often **a group of nodes lies on the shortest paths between other nodes**.

A group with high betweenness can **control information flow in the network**.

Equation

$$C_B(G) = \sum_{s \neq t} \frac{\sigma_{st}(G)}{\sigma_{st}}$$

Where:

- (σ_{st}) = total shortest paths between nodes (s) and (t)
- $\sigma_{st}(G)$ = number of shortest paths passing through group (G)

3. Group Closeness Centrality

It measures **how close a group of nodes is to all other nodes in the network**.

Equation

$$C_c(G) = \frac{1}{\sum d(G, v)}$$

Where:

- $d(G, v)$ = shortest distance between group (G) and node (v)

Example

Consider a social network where a **group of influencers** frequently interact with many other users.

This group can **spread information quickly** and influence opinions.

On platforms like Facebook or Instagram, a **group of popular content creators** may have high **group centrality** because they collectively reach a large audience.

Applications of Group Centrality

- Identifying **influential communities**
- Studying **information diffusion**
- Detecting **key teams in organizations**
- Analyzing **collaboration networks**

✓ Summary:

Group centrality measures the **importance of a group of nodes in a network** rather than individual nodes. It helps understand how **groups influence communication, control information flow, and interact with the rest of the network.**

Herd Behavior

Ans. **Herd Behavior**

Herd Behavior refers to the phenomenon where **individuals follow the actions or decisions of a larger group**, often ignoring their own information or personal judgment.

In **social computing and social networks**, people tend to **imitate the behavior of others**, especially when they are uncertain about what decision to make.

Definition

Herd behavior is the tendency of individuals to **copy the actions of a group**, assuming that the group's decision is correct.

It occurs because people believe that **if many others are doing something, it must be the right choice.**

Example

Suppose a video becomes popular on a social media platform:

1. A few people start watching and liking the video.
2. Many others see the high number of likes and views.
3. They also start watching and sharing it.

Soon the video becomes **viral** because people follow the **crowd behavior**.

This is common on platforms like YouTube, Instagram, and Twitter.

Causes of Herd Behavior

1. Social Influence

People follow others to fit into a group.

2. Information Cascades

Individuals assume others have better information.

3. Uncertainty

When people are unsure, they rely on the decisions of others.

4. Peer Pressure

Desire to conform to group behavior.

Characteristics of Herd Behavior

- People **imitate others' actions**
- Decisions are based on **group behavior**
- Can spread **rapidly in networks**
- Often leads to **trends or viral content**

Applications and Examples

- **Viral trends on social media**
- **Stock market bubbles**
- **Online product popularity**

- **Fashion trends**

Example: When many people buy a product after seeing positive reviews on Amazon, others also buy it due to **herd behavior**.

✅ Summary:

Herd behavior occurs when individuals **follow the actions of a group rather than making independent decisions**. It plays an important role in **trend formation, viral content, and collective behavior in social networks**.

Epidemics

Ans. **Epidemics**

In **social computing and network science**, **epidemics** refer to the **spread of diseases, information, behaviors, or viruses through a network of individuals**. The concept is borrowed from epidemiology, where it describes how **infectious diseases spread among people**.

In social networks, epidemics can also represent the spread of:

- **Rumors**
- **Ideas**
- **Computer viruses**
- **Viral content**

Definition

An **epidemic** is the **rapid spread of a disease or information through a population or network over a period of time**.

In network terms:

- **Nodes** represent individuals.
- **Edges** represent interactions or connections through which the disease or information spreads.

Example

Suppose one person in a group is infected with a virus.

1. Person **A** is infected.
2. A interacts with **B and C**.
3. B and C become infected.
4. B and C spread the infection to others.

Thus, the disease spreads through the network like a **cascade**.

Similarly, a viral post on platforms like Facebook or Twitter spreads in a similar way.

Epidemic Models

1. SI Model (Susceptible–Infected)

Individuals can be:

- **Susceptible (S)** – not infected but can become infected
- **Infected (I)** – currently infected

Once infected, the individual **remains infected permanently**.

2. SIS Model (Susceptible–Infected–Susceptible)

States:

- **Susceptible (S)**
- **Infected (I)**

Infected individuals can **recover and become susceptible again**.

Example: Some diseases where immunity is temporary.

3. SIR Model (Susceptible–Infected–Recovered)

States:

- **Susceptible (S)**
- **Infected (I)**
- **Recovered (R)**

Recovered individuals **gain immunity and cannot be infected again**.

Example: Diseases like COVID-19 or measles.

Applications of Epidemic Models

- Studying **disease spread**
- Understanding **information diffusion in social networks**
- Modeling **computer virus propagation**
- Predicting **viral marketing success**

Characteristics of Epidemics

- Spread through **network connections**
- Can grow **rapidly**
- Depends on **interaction patterns**
- Influenced by **network structure**

✓ **Summary:**

An **epidemic** describes the **spread of disease, information, or behavior through a network of individuals**. Epidemic models such as **SI, SIS, and SIR** help researchers understand and predict how such spreading processes occur in social and communication networks.